Text-to-Video Generation using Diffusion Models: A Structured Analysis of Human Motion in Dynamic Environments

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***Abstract*—Text-to-video generation has recently gained significant attention with the advancement of diffusion-based generative models. While these models have achieved remarkable success in image synthesis, extending them to video generation introduces additional challenges related to temporal consistency and motion realism.**

In this work, we propose a two-stage diffusion-based pipeline that combines Stable Diffusion XL for text-to-image generation and Stable Video Diffusion for image-to-video synthesis. The system is evaluated under a unified theme focusing on human motion in dynamic environments, including crowded streets and public spaces.

We conduct structured experiments by varying guidance scale, inference steps, and scene complexity to analyze their impact on video quality and consistency. The results demonstrate that while the model maintains strong semantic alignment with input prompts, it struggles with temporal consistency and realistic motion in complex scenarios.

Our findings highlight key limitations of current diffusion-based video generation models and provide insights for future improvements in temporal modeling and robustness.

In Phase 2, we further improve the baseline pipeline by introducing two enhancement strategies: semantic prompt conditioning and temporal stabilization. An ablation study is conducted to compare the baseline, each enhancement independently, and their combined effect. The best enhanced model improves mean SSIM from 0.7049 to 0.9567 and extends the final video duration to 4.08 seconds, satisfying the final project requirement.

I. Introduction

Diffusion models have recently achieved remarkable success in generative tasks, particularly in high-quality image synthesis. Extending these models to video generation introduces additional complexity due to the temporal dimension, where maintaining consistency across frames becomes a fundamental challenge.

In this project, we develop a text-to-video generation pipeline based on diffusion models, aiming to generate short, temporally coherent video clips from natural language descriptions. Unlike image generation, video synthesis requires the model to capture both spatial details and temporal dynamics, making the task significantly more challenging.

A. Motivation

This work focuses on generating **human-centered dynamic scenes in complex environments**, such as crowded streets and public markets. These scenarios are deliberately selected because they involve complex motion patterns, multiple interacting entities, and frequent occlusions, which expose common weaknesses in current video generation models.

The primary motivation is to evaluate the model's performance under challenging conditions, with emphasis on:

- Temporal consistency across frames
- Realism of human motion
- Semantic alignment between text prompts and generated videos

These aspects are critical for producing realistic and coherent video outputs, especially in scenes with high motion complexity.

B. Problem Statement

Despite recent advancements, diffusion-based video generation models still suffer from several limitations. These include difficulties in maintaining consistent object structure over time, generating realistic motion, and preserving alignment with the input text in complex scenes.

This project aims to systematically analyze these limitations through structured experiments using a controlled text-to-video pipeline.

C. Contributions

The contributions of this work can be summarized as follows:

- We design and implement a text-to-video generation pipeline combining Stable Diffusion XL (text-to-image) and Stable Video Diffusion (image-to-video)
- We introduce a theme-driven evaluation framework focusing on human motion in dynamic environments
- We conduct structured experiments by varying guidance scale, inference steps, and scene complexity
- We perform quantitative and heuristic-based weakness analysis using metrics such as CLIP-SIM and SSIM

- We implement Phase 2 enhancements including semantic prompt conditioning, temporal stabilization, and an ablation study comparing baseline, Enhancement A, Enhancement B, and their combination
- We add an optional neural text-to-audio module using SpeechT5 to generate synchronized narration for the final video

II. Related Work

The field of generative modeling has witnessed rapid advancements in recent years, particularly with the introduction of diffusion-based models. These models have demonstrated superior performance in image synthesis tasks and have recently been extended to video generation. In this section, we review the most relevant work in diffusion models, text-to-image generation, and text-to-video synthesis.

A. Diffusion Models

Diffusion models were introduced as a probabilistic framework for generative modeling by Ho et al., where data generation is formulated as the reverse of a gradual noise corruption process. These models, commonly referred to as Denoising Diffusion Probabilistic Models (DDPM), have shown strong performance in generating high-quality images while maintaining stable training dynamics.

Subsequent work by Song et al. introduced Denoising Diffusion Implicit Models (DDIM), which improved sampling efficiency by reducing the number of required diffusion steps. These improvements made diffusion models more practical for real-world applications.

Furthermore, latent diffusion models significantly enhanced scalability by performing the diffusion process in a compressed latent space rather than the pixel space. This approach reduces computational complexity while maintaining high output quality, enabling the generation of high-resolution images.

B. Text-to-Image Generation

Text-to-image generation has been a major research area, with models such as DALL-E, GLIDE, and Stable Diffusion demonstrating the ability to generate realistic images from natural language descriptions. Stable Diffusion XL (SDXL) represents an advanced version of latent diffusion models, offering improved image quality, better prompt understanding, and enhanced compositional capabilities.

These models rely on cross-attention mechanisms to incorporate textual information into the generation process, allowing them to align visual outputs with semantic descriptions effectively.

C. Text-to-Video Generation

Extending generative models from images to videos introduces additional challenges due to the temporal dimension. Several approaches have been proposed to address this problem.

Imagen Video extends diffusion models to video generation by progressively generating frames at multiple resolutions, achieving high-quality results but requiring significant computational resources. Similarly, Make-A-Video introduces a method for generating videos from text without requiring large-scale paired video-text datasets.

Phenaki proposes a transformer-based approach capable of generating long videos from textual input, demonstrating the potential for open-domain video generation.

Stable Video Diffusion (SVD) builds upon latent diffusion techniques and introduces temporal modeling mechanisms to generate short video clips from images. By conditioning on an initial frame, the model is able to maintain a degree of consistency across generated frames.

D. Evaluation Metrics for Video Generation

Evaluating generative video models remains a challenging task. Several metrics have been proposed to measure different aspects of video quality.

Frechet Video Distance (FVD) is commonly used to evaluate the realism of generated videos by comparing feature distributions between generated and real samples. Structural Similarity Index (SSIM) measures the similarity between consecutive frames and is often used to assess temporal consistency.

CLIP-based similarity metrics have also gained popularity for evaluating semantic alignment between generated images or frames and textual prompts.

E. Limitations of Existing Work

Despite significant progress, current text-to-video generation models still face several limitations. Maintaining temporal consistency across frames remains a major challenge, particularly in scenes with complex motion and multiple interacting entities.

Additionally, generating realistic human motion is difficult due to the variability and complexity of human behavior. Models often produce artifacts such as flickering, inconsistent object structures, and unnatural motion patterns.

This work aims to build upon existing research by providing a structured experimental analysis of these limitations, focusing specifically on human-centered dynamic environments.

III. Mathematical Formulation

Diffusion models are based on a probabilistic framework that models data generation as a gradual denoising process. This section presents the mathematical foundations underlying the diffusion process used in this work.

A. Forward Diffusion Process

The forward process gradually adds Gaussian noise to the original data over a sequence of timesteps. Given an original sample x_0, the noisy version at timestep t is defined as:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I) \tag{1}$$

where β_t represents the variance schedule controlling the amount of noise added at each step.

This process can be expressed in closed form as:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon \tag{2}$$

where $\epsilon \sim \mathcal{N}(0, I)$ and $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

B. Reverse Diffusion Process

The reverse process aims to reconstruct the original data by removing noise step-by-step. A neural network parameterized by θ is trained to approximate:

$$p_\theta(x_{t-1} \mid x_t) \tag{3}$$

This is achieved by predicting the noise component added during the forward process.

C. Training Objective

The model is trained using a simplified objective derived from the Evidence Lower Bound (ELBO), which reduces to a Mean Squared Error (MSE) loss:

$$\mathcal{L} = E_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right] \tag{4}$$

This objective enables the model to learn how to effectively remove noise at each timestep.

D. Classifier-Free Guidance

To improve the alignment between generated samples and input text, classifier-free guidance is applied during inference:

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, c) + w \cdot (\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \emptyset)) \tag{5}$$

where w is the guidance scale controlling the strength of conditioning.

E. Temporal Consistency Metrics

To evaluate temporal consistency in generated videos, we use Structural Similarity Index (SSIM):

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{6}$$

where μ and σ represent mean and variance statistics of image patches.

F. Semantic Similarity (CLIP)

Semantic alignment is evaluated using CLIP-based similarity, which measures the cosine similarity between image and text embeddings:

$$\text{CLIP-SIM} = \frac{f_{img}(x) \cdot f_{text}(t)}{\|f_{img}(x)\| \|f_{text}(t)\|} \tag{7}$$

This metric quantifies how well the generated frames match the input textual description.

IV. Methodology

In this work, we propose a structured text-to-video generation pipeline based on diffusion models. The methodology is designed to systematically analyze the performance of the model under controlled experimental conditions, with a particular focus on human motion in dynamic environments.

A. Pipeline Overview

The proposed system follows a two-stage generation pipeline:

1) Text-to-Image Generation using Stable Diffusion XL (SDXL)
2) Image-to-Video Generation using Stable Video Diffusion (SVD)

In the first stage, a textual prompt is converted into a high-quality image that captures the spatial structure and semantic content of the scene. In the second stage, this image is used as input to a video diffusion model, which generates a sequence of frames to form a short video clip.

This design allows us to decouple spatial and temporal generation, improving stability and enabling more controlled experimentation.

B. Text-to-Image Generation (SDXL)

Stable Diffusion XL is a latent diffusion model that generates images conditioned on textual input. The model operates in a compressed latent space using a Variational Autoencoder (VAE), which significantly reduces computational cost.

The generation process is guided by a text encoder that transforms the input prompt into a latent representation. This representation is integrated into the diffusion model using cross-attention mechanisms, enabling the model to align the generated image with the textual description.

The output of this stage serves as the initial frame for video generation.

C. Image-to-Video Generation (SVD)

Stable Video Diffusion extends diffusion models to the temporal domain by generating multiple frames from a single input image. The model incorporates temporal modeling techniques to maintain consistency across frames.

Unlike frame-by-frame generation approaches, SVD processes temporal information jointly, allowing it to capture motion dynamics and reduce flickering artifacts.

Key parameters used in this stage include:

- Number of inference steps
- Motion bucket ID (controls motion intensity)
- Noise augmentation strength

These parameters directly influence the quality and realism of the generated video.

D. Theme-Driven Experiment Design

All experiments in this work are conducted under a unified theme: **human motion in dynamic environments**. This includes scenarios such as walking individuals, busy streets, and crowded public spaces.

To systematically evaluate model performance, we define three levels of scene complexity:

- **Simple:** A single person walking in an empty environment
- **Medium:** A person walking in a moderately busy street with background motion

- **Complex:** A crowded environment with multiple interacting individuals

This structured approach allows us to analyze how model performance degrades as scene complexity increases.

E. Experimental Configurations

We conduct three main experiments to analyze the impact of key parameters:

1) Experiment A: Guidance Scale Variation: In this experiment, we fix the prompt and vary the guidance scale parameter. This parameter controls the strength of conditioning on the input text.

Higher guidance values improve semantic alignment but may reduce diversity and introduce artifacts.

2) Experiment B: Inference Steps Variation: In this experiment, we vary the number of inference steps used in the video generation stage.

Increasing the number of steps generally improves output quality and reduces noise, but at the cost of higher computational time.

3) Experiment C: Scene Complexity Analysis: This experiment evaluates the model across different levels of scene complexity (simple, medium, complex) to analyze its robustness in increasingly challenging scenarios.

F. Evaluation Metrics

To evaluate the generated videos, we use a combination of quantitative and heuristic metrics:

- **CLIP-SIM:** Measures semantic alignment between the input prompt and generated frames
- **SSIM:** Measures structural similarity between consecutive frames to assess temporal consistency
- **Frame Difference:** Measures pixel-level variation across frames

These metrics provide insights into both visual quality and temporal coherence.

G. Weakness Analysis Framework

In addition to quantitative metrics, we implement a heuristic-based analysis framework to identify model weaknesses.

The system automatically categorizes results based on:

- Temporal consistency (good, moderate, poor)
- Flickering detection
- Semantic alignment quality

This approach allows for a structured interpretation of results and helps highlight key limitations of the model.

H. Implementation Details

The entire pipeline is implemented in Python using the HuggingFace Diffusers library. PyTorch is used as the primary deep learning framework, with GPU acceleration enabled when available.

The system is designed to be modular, with separate components for:

- Model loading
- Video generation
- Evaluation metrics
- Experiment execution

All outputs, including frames, videos, and metadata, are saved in a structured directory format to ensure reproducibility and facilitate analysis.

V. Results

In this section, we present a comprehensive analysis of the experimental results obtained from the proposed text-to-video generation pipeline. The evaluation focuses on three main aspects: semantic alignment, temporal consistency, and robustness under varying levels of scene complexity.

The results are derived from a structured set of experiments designed to isolate the effect of key parameters, including guidance scale, inference steps, and scene complexity.

A. Overall Performance Analysis

Across all experiments, the model demonstrates a strong ability to generate visually plausible frames that align well with the input textual prompts. This is reflected in consistently high CLIP-SIM scores, indicating effective semantic conditioning.

However, temporal consistency remains a significant challenge. While short sequences exhibit reasonable coherence, longer or more complex sequences tend to suffer from instability, including flickering and structural inconsistencies.

These findings suggest that while the model is effective in capturing spatial features, it struggles to maintain temporal relationships across frames.

B. Experiment A: Effect of Guidance Scale

The results of Experiment A show a clear relationship between guidance scale and semantic alignment. As the guidance scale increases, the model produces outputs that more closely match the textual prompt, resulting in higher CLIP-SIM scores.

However, this improvement comes with trade-offs. At higher guidance values, the model tends to produce over-constrained outputs, leading to reduced diversity and occasional visual artifacts. In some cases, excessive guidance results in unnatural textures and overly sharp transitions between frames.

At lower guidance values, the outputs are more diverse but less aligned with the prompt, indicating a weaker conditioning signal.

This highlights the importance of selecting an optimal guidance scale that balances alignment and diversity.

C. Experiment B: Effect of Inference Steps

Increasing the number of inference steps generally leads to improvements in visual quality and temporal smoothness. Videos generated with higher step counts exhibit reduced noise and more stable frame transitions.

However, the improvement is not linear. Beyond a certain threshold, additional steps result in diminishing returns, where the increase in quality is marginal compared to the additional computational cost.

This observation suggests that there exists an optimal range of inference steps that balances efficiency and quality.

D. Experiment C: Impact of Scene Complexity

Scene complexity has the most significant impact on model performance. In simple scenarios, the model produces stable and coherent outputs with minimal artifacts. The generated motion appears smooth, and object structures are well preserved.

In medium complexity scenarios, minor inconsistencies begin to appear. These include slight distortions in object shapes and occasional flickering between frames.

In complex scenes, the model struggles significantly. The presence of multiple interacting entities introduces challenges that the model is unable to handle effectively. This results in:

- Inconsistent object shapes across frames
- Noticeable flickering and instability
- Unrealistic motion patterns

These issues indicate that the model has limited capacity to model complex temporal interactions.

E. Temporal Consistency Analysis

Temporal consistency is evaluated using SSIM and frame difference metrics. Higher SSIM values indicate stronger similarity between consecutive frames, suggesting stable motion.

In simple scenes, SSIM values are relatively high, indicating consistent frame transitions. In contrast, complex scenes show lower SSIM values and higher frame differences, reflecting instability and flickering.

Variance in SSIM scores further highlights fluctuations in temporal consistency, particularly in scenes with dynamic interactions.

F. Semantic Alignment Analysis

Semantic alignment is evaluated using CLIP-SIM scores. The results show that the model maintains strong alignment across most configurations, particularly at higher guidance scales.

However, in complex scenes, alignment may degrade slightly due to the increased difficulty of representing multiple entities and interactions accurately.

G. Weakness Analysis

Based on both quantitative metrics and visual inspection, several key weaknesses are identified:

- **Temporal Instability:** The model fails to maintain consistent object structure across frames, leading to flickering artifacts.
- **Motion Unrealism:** Generated motion often lacks physical plausibility, especially in crowded environments.
- **Scalability Issues:** Performance degrades significantly as scene complexity increases.
- **Parameter Sensitivity:** Output quality is highly sensitive to parameter selection, requiring careful tuning.

VI. Phase 2 Enhancements and Ablation Study

Based on the Phase 1 weakness analysis, temporal consistency was identified as the main limitation of the baseline system. Although the generated videos achieved strong semantic alignment according to CLIP-SIM, several outputs suffered from flickering, unstable object boundaries, and inconsistent motion across frames. Therefore, Phase 2 focuses on improving temporal coherence while maintaining semantic alignment.

A. Enhancement A: Semantic Prompt Conditioning

The first enhancement targets text-video semantic alignment. We improve the original prompt by adding explicit constraints related to human anatomy, consistent clothing, natural walking motion, stable camera movement, and realistic lighting. A negative prompt is also introduced to reduce common artifacts such as flickering, distorted bodies, extra limbs, unstable backgrounds, and low-quality visual details.

In addition, the classifier-free guidance scale is increased from 7.5 to 10.0. Classifier-free guidance modifies the denoising prediction as:

$$\hat{\epsilon}_\theta = \epsilon_\theta(x_t, \emptyset) + w\left(\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t, \emptyset)\right) \tag{8}$$

where c is the text condition and w is the guidance scale. Increasing w strengthens the influence of the textual condition, which can improve prompt alignment, but may reduce diversity or introduce artifacts if set too high.

B. Enhancement B: Temporal Stabilization

The second enhancement directly addresses temporal inconsistency. We reduce the motion intensity and noise augmentation parameters in Stable Video Diffusion by lowering the motion bucket ID and noise augmentation strength. This reduces aggressive motion changes between frames.

A lightweight temporal smoothing post-process is then applied:

$$\tilde{I}_t = \alpha I_t + (1-\alpha)\tilde{I}_{t-1} \tag{9}$$

where I_t is the generated frame at time t, $\tilde{I}_t$ is the smoothed frame, and α controls the trade-off between preserving motion and suppressing flicker. In our implementation, this smoothing is followed by linear frame interpolation to increase the final video duration beyond four seconds while producing smoother transitions.

C. Combined Enhancement

The combined configuration applies both semantic prompt conditioning and temporal stabilization. This variant tests whether stronger semantic control and temporal smoothing can improve the generated video jointly.

D. Ablation Study

To evaluate the effectiveness of the proposed enhancements, we compare four variants:

- **Baseline:** Original Phase 1 pipeline
- **Enhancement A:** Prompt conditioning and stronger CFG
- **Enhancement B:** Temporal stabilization and interpolation
- **Combined A+B:** Both enhancements together

The ablation results show that Enhancement B provides the strongest overall improvement. Compared with the baseline, it increases mean SSIM from 0.7049 to 0.9567 and reduces mean frame difference from 19.9993 to 4.4957. It also achieves the highest CLIP-SIM score of 34.9508, indicating that temporal stabilization does not harm semantic alignment.

The combined variant achieves the lowest frame difference and highest PSNR, but its CLIP-SIM score is slightly lower than Enhancement B. Therefore, Enhancement B is selected as the final video configuration because it provides the best balance between semantic alignment, temporal consistency, and the required final duration.

E. Final Video Selection

The final selected video is generated using Enhancement B. It has a duration of 4.08 seconds, satisfying the Phase 2 requirement of producing a final generated video of at least four seconds. The final output demonstrates smoother motion, reduced flickering, and stronger temporal consistency compared with the Phase 1 baseline.

F. Optional Bonus: Neural Text-to-Audio

As an optional bonus, we integrate a neural text-to-audio module with the final generated video. The module uses SpeechT5 to synthesize narration from a short textual description of the scene. The generated audio is then synchronized and merged with the final enhanced video using FFmpeg.

The bonus module includes multiple advanced features:

- Neural TTS model integration using SpeechT5
- Multi-voice support through different speaker embeddings
- Style and emotion controls such as neutral, happy, calm, and clear speech
- Adjustable speed and pitch parameters
- Context-aware pauses based on punctuation

This component improves the storytelling quality of the final demo and demonstrates an additional multimodal extension beyond basic text-to-video generation.

G. Discussion

The results highlight a fundamental limitation of current diffusion-based video models: while they excel at generating high-quality individual frames, they struggle to maintain coherence across time.

This limitation becomes more pronounced in scenarios involving multiple moving objects and complex interactions. The lack of explicit temporal constraints in the training objective contributes to these issues.

Overall, the findings emphasize the need for improved temporal modeling techniques and more robust training strategies for video generation.

VII. Conclusion

In this work, we presented a comprehensive study of diffusion-based text-to-video generation using a two-stage pipeline that combines Stable Diffusion XL for image synthesis and Stable Video Diffusion for temporal generation. In Phase 1, we analyzed the baseline system under different guidance scales, inference steps, and scene complexity levels. In Phase 2, we implemented targeted enhancements to address the identified weaknesses, including semantic prompt conditioning and temporal stabilization.

Through a series of structured experiments, we analyzed the impact of key parameters, including guidance scale, inference steps, and scene complexity, on the quality of generated videos. The results demonstrate that diffusion-based models are highly effective in producing visually plausible frames with strong semantic alignment to textual prompts.

However, despite these strengths, the model exhibits notable limitations in maintaining temporal consistency and generating realistic motion, particularly in complex scenes involving multiple interacting entities. As scene complexity increases, the quality of the generated videos degrades significantly, leading to artifacts such as flickering, inconsistent object structures, and unrealistic motion patterns.

The experimental findings highlight a fundamental gap between spatial generation capabilities and temporal modeling in current diffusion-based systems. While the models excel at generating individual frames, they struggle to preserve coherence across time, which is essential for high-quality video generation.

Overall, this work provides a structured and in-depth analysis of diffusion-based video generation systems, emphasizing both their potential and their current limitations. These insights contribute to a better understanding of the challenges in this field and serve as a foundation for future research.

VIII. Future Work

The findings of this work reveal several opportunities for improving diffusion-based video generation systems. Future research can focus on addressing the identified limitations, particularly in temporal consistency and motion realism.

One promising direction is the incorporation of explicit temporal consistency constraints into the training objective. Current models rely primarily on spatial denoising objectives, which do not directly enforce coherence across frames. Introducing temporal loss functions could help reduce flickering and improve stability.

Another potential improvement involves enhancing temporal modeling architectures. This includes the use of more advanced temporal attention mechanisms, 3D convolutional networks, or hybrid approaches that combine diffusion models with sequence modeling techniques such as transformers.

TABLE I
PHASE 2 ABLATION STUDY RESULTS

Variant	CLIP-SIM	Mean SSIM	Mean PSNR	Frame Diff.	Duration (s)	Status
Baseline	34.2195	0.7049	16.4532	19.9993	3.57	Moderate
Enhancement A	33.8013	0.8064	20.0771	10.1719	3.57	Moderate
Enhancement B	34.9508	0.9567	27.9292	4.4957	4.08	Good
Combined A+B	33.6461	0.9466	28.4856	3.9539	4.08	Good

Additionally, incorporating motion-aware conditioning mechanisms could improve the realism of generated motion. For example, integrating optical flow or pose estimation signals may help guide the model in generating more physically plausible movements.

Evaluation methods can also be improved by incorporating more robust metrics such as Frechet Video Distance (FVD), which requires comparison against real video datasets. This would provide a more accurate assessment of video realism.

Furthermore, optimizing the efficiency of the generation process is an important area for future work. Diffusion models are computationally expensive, and reducing inference time while maintaining quality would make them more practical for real-world applications.

Finally, expanding the scope of experiments to include longer video sequences and more diverse scenarios would provide deeper insights into model performance and scalability.

By addressing these challenges, future research can significantly enhance the capabilities of text-to-video generation systems and move closer to achieving realistic and temporally coherent video synthesis.

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